

Bhavani Shankar Y

Applied Scientist – NLP — LLM Systems — Agent Architectures

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RESEARCH SUMMARY

Applied AI researcher with 12+ years of experience designing and deploying machine learning systems at scale. Specialized in Large Language Models (LLMs), planning-based agent architectures, structured workflow synthesis, and knowledge graph construction. Proven ability to translate research prototypes into production-grade distributed systems with measurable business impact.

TECHNICAL EXPERTISE

- **LLM Systems:** Fine-tuning, Structured Output Control, Plan-Execute Architectures, Tool-Augmented Agents, Hallucination Mitigation, RAG.
- **ML Algorithms:** Clustering, Classification, Regression, NLP, Feature Engineering.
- **Agent Frameworks:** LangGraph, Planner-Executor Patterns, Tool Routing.
- **Programming & Data:** Python, Java, Go; Postgres, Qdrant, Cassandra, Elasticsearch, MongoDB.
- **Infrastructure:** AWS (EKS), Azure, Kubernetes, Docker, Temporal, Kafka, SQS.

EXPERIENCE

FourKites India Pvt. Ltd

2018 – Present

Senior Engineering Manager – Applied AI Systems (2023–Present)

- Designed and deployed LLM-based systems for supply chain automation, enabling structured query generation and tool-based execution.
- Improved ETA prediction accuracy by 15% through model refinement and feature engineering across multi-modal transportation data.
- Led development of conversational and document intelligence agents with focus on cost-performance optimization and system reliability.
- Conducted internal evaluation studies on planning-based architectures to improve workflow stability and execution consistency.

Staff Software Engineer (2020–2023)

- Architected cloud-native migration from on-premise to Azure, resulting in \$500K annual cost savings.
- Designed distributed, event-driven ML service pipelines using Kubernetes and automated CI/CD workflows.
- Guest speaker at the 6th World API and Developer Conference.

Associate Engineering Manager (2018–2020)

- Built and scaled mobile-native engineering team; improved Play Store rating from 2.7 to 4.6.
- Designed end-to-end "Courier Mode" tracking system from ingestion to automated customer notification.

Samsung R&D India, Bangalore

2013 – 2018

Technical Lead – On-device ML Research

- Led research on unsupervised clustering algorithms for automated on-device knowledge graph generation from web content.
- Designed graph-based personalization systems deployed across Samsung Galaxy flagship devices.
- Investigated optimal cluster estimation and automatic cluster naming strategies.
- Commercialized "Smart Stories" media clustering and "Keyword Suggestion" features in Bixby Shopping.

PROJECTS

Loft – Natural Language to Executable Workflow Compiler

- Designed domain-specific LLM-powered workflow synthesis system converting natural language into deterministic DAG-based execution plans.

- Implemented planner-executor architecture using LangGraph for multi-step task decomposition.
- Developed structured intermediate representations to ensure reproducible and stable workflow generation.
- Integrated Qdrant (vector retrieval) and Postgres (short-term memory) enabling adaptive re-planning and hallucination mitigation.
- Deployed asynchronous execution pipeline using Temporal on AWS EKS.
- Reduced enterprise integration time from 4 weeks to <0.5 days in internal evaluation.

Document Intelligence Agent

- Designed LLM-powered document processing system for structured entity extraction from supply chain documents (BOL, invoices, shipment updates).
- Built human-in-the-loop (HITL) validation framework to ensure extraction precision and reduce downstream propagation errors.
- Evaluated AI-native extraction strategies versus OCR-based pipelines across cloud providers.
- Designed feedback loops and correction workflows to improve extraction consistency.
- Successfully onboarded 10 pilot customers demonstrating measurable operational efficiency gains.

Conversational AI Assistant – Structured Data Reasoning

- Architected LLM-based conversational interface enabling natural language interaction over structured enterprise data.
- Implemented tool-augmented execution layer to translate user queries into filter, aggregation, and action-triggering workflows.
- Conducted fine-tuning and prompt optimization experiments to improve task execution accuracy and reduce hallucinations.
- Designed streaming and non-streaming response workflows for low-latency inference.
- Integrated AI Gateway architecture for model routing, observability, and reliability.

PUBLICATIONS, PATENTS & AWARDS

- **Patent:** “Dynamically modifying elements of user interface based on knowledge graph,” WO 2015122691.
- **Paper:** “Study of security issues in cloud computing,” ICJST, Vol. 2.
- **SBPA 2018:** Unsupervised clustering algorithm for community detection using media.
- **SBPA 2014:** Novel knowledge graph construction using browser history.
- **Winner:** Company-wide hackathon, Illuminati’20.

EDUCATION

- **PG Level Certification:** AI and Machine Learning, IIIT Hyderabad (2022).
- **M.Tech:** Computer Science Engineering, IIIT Hyderabad (2013).
- **B.Tech:** Computer Science Engineering, V.R. Siddhartha College of Engineering (2011).